VoiceBridge – Multilingual AI Voice Assistant - https://github.com/tsar0705/VoiceBridge

Project by: Anantharaman T S
Institution: Amrita Vishwa Vidyapeetham – Amrita School of AI, Coimbatore

The problem

Many people in India are more comfortable speaking than typing, especially when communicating in their native language. Conventional digital assistants can struggle with regional-language speech, code-mixed expressions, and natural voice interactions. This creates friction for users who want quick, accessible information through conversation.

Our solution

VoiceBridge is a voice-first AI assistant designed for multilingual Indian users. Users can speak naturally, receive a transcription, interact with an AI assistant, and listen to the response as generated speech.

The application supports a conversational workflow with Tamil and English tested end to end. It also includes language and response-mode options for additional Indian-language and code-mixed interactions.

How it works

1. **Voice capture:** The React frontend records speech through the browser microphone.
2. **Audio processing:** Browser-recorded audio is converted into genuine 16 kHz, mono, 16-bit PCM WAV to ensure compatibility with the speech recognition API.
3. **Speech recognition:** Gnani AI Prisma converts the audio into text and detects the spoken language.
4. **AI reasoning:** Google Gemini generates a context-aware response using the conversation history.
5. **Speech synthesis:** Gnani AI Timbre converts the response into audio.
6. **Voice interaction:** The frontend displays the transcript and response and plays the generated speech. Conversation context is retained across turns and can be reset.

Technology used

- **Gnani AI Prisma v3.0:** Speech-to-text transcription.
- **Gnani AI Timbre v2.5:** Text-to-speech synthesis.
- **Google Gemini 3.5 Flash Lite:** Conversational response generation.
- **Frontend:** React, Vite, JavaScript, Tailwind CSS and browser Web Audio APIs.
- **Backend:** Python, FastAPI and HTTPX.

What has been built and tested

The working application has completed Tamil and English speech-to-text, AI response generation and text-to-speech flows. Browser-driven tests verified PCM WAV conversion, multipart audio upload, multi-turn conversation continuity, audio playback and session reset.

The backend test suite passed 67 tests, and the frontend production build completed successfully.

Why it matters

VoiceBridge explores how Indian speech technologies can make AI interaction more accessible and natural. By combining Gnani's speech models with conversational reasoning, it demonstrates a practical voice-first experience that can be extended to more Indian languages, code-mixed conversations and real-world use cases.

Project contribution: Independently designed and implemented the application architecture, frontend, backend, Gnani API integrations, audio conversion pipeline, conversation management and testing.